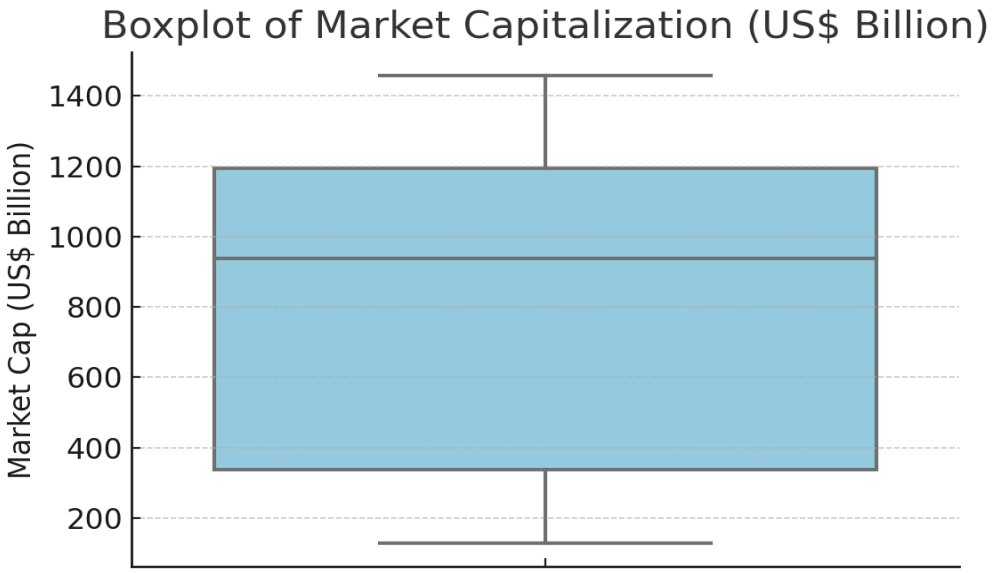
ETL Bank Data Analysis

1. Data Preparation

The dataset was extracted from an archived Wikipedia page listing the largest banks globally. PySpark was used to initialize a session and ingest the table, which was converted into a DataFrame. Column names were standardized (lowercase, underscores) and the 'market_cap' column was parsed into numeric values (US$ billions).

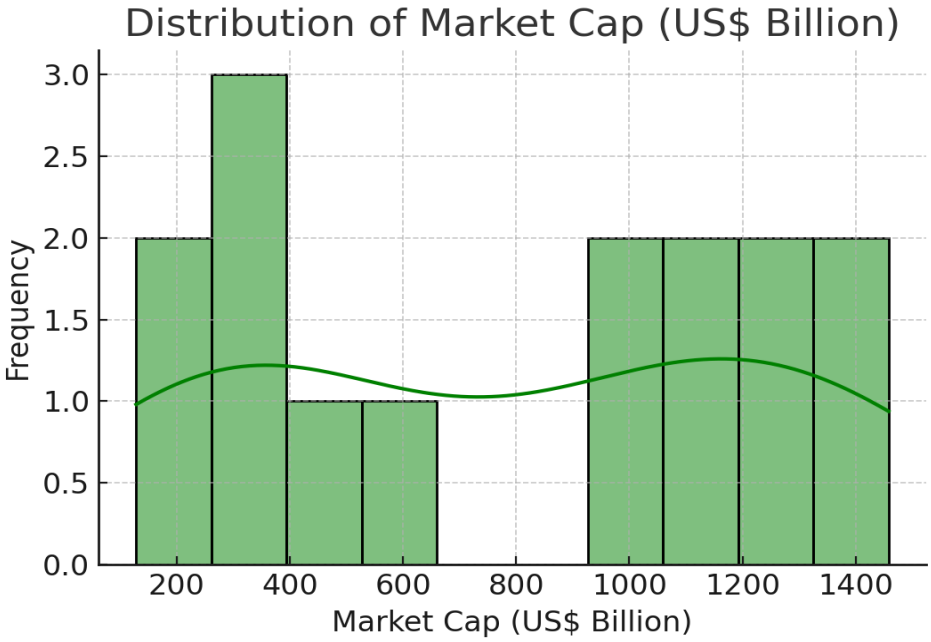
2. Data Cleaning

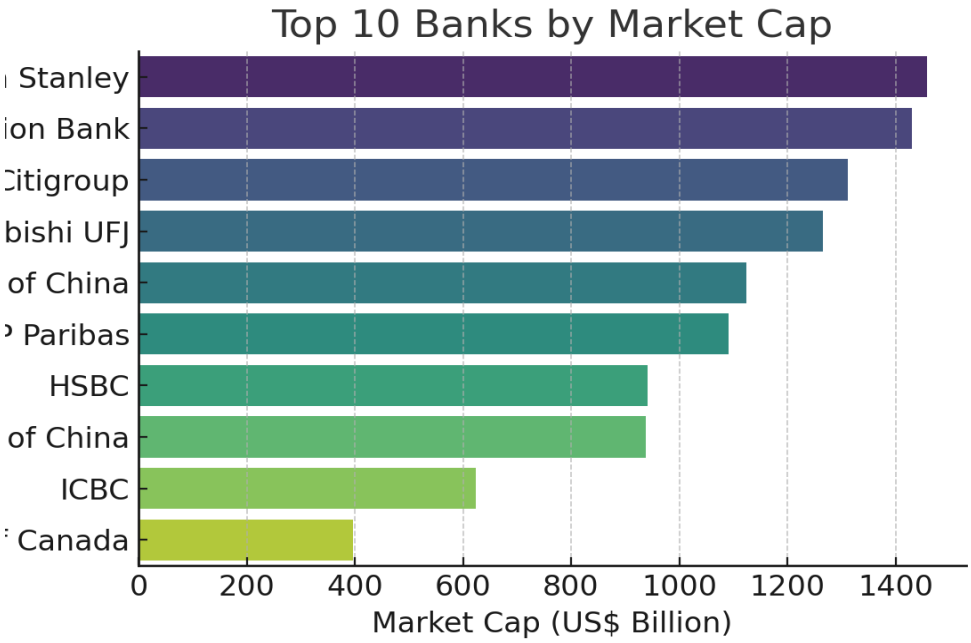
Rows with missing or null values were dropped using PySpark's DataFrame operations. Market capitalization values were cleaned to remove commas, currency symbols, and converted to billions. Final cleaned dataset ensured numeric consistency across all records.

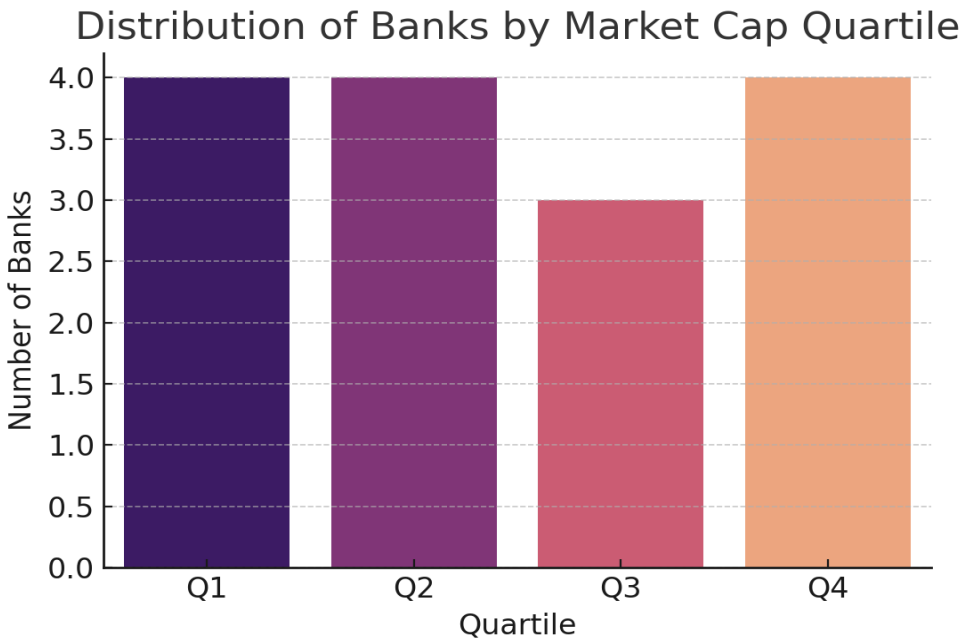


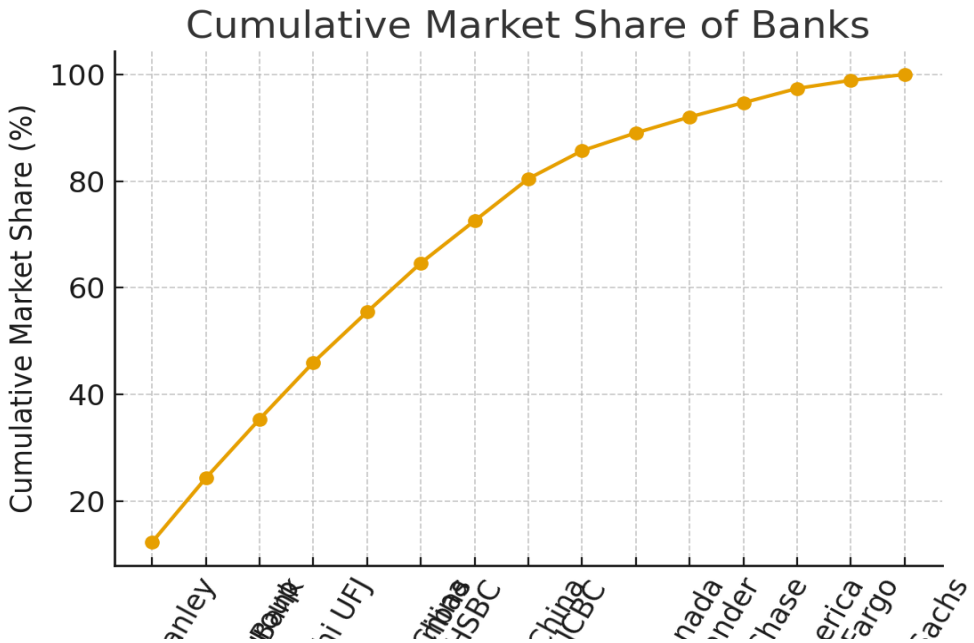
3. Exploratory Data Analysis

Several analyses were carried out to understand the distribution of market capitalization across banks. This included histograms, top 10 banks ranking, quartile-based segmentation, and market dominance plots.


Distribution of Market Cap (US$ Billion)
Frequency
3.0
2.5
2.0
1.5
1.0
0.5
0.0
200
400
600
800
1000
1200
1400
Market Cap (US$ Billion)


Top 10 Banks by Market Cap
Stanley
ion Bank
Citigroup
bishi UFJ
of China
Paribas
HSBC
of China
ICBC
Canada
0
200
400
600
800
1000
1200
1400
Market Cap (US$ Billion)





Conclusion

The analysis highlights the distribution of market capitalization across leading banks globally. Using PySpark for ETL and Pandas/Seaborn for visualization, we identified outliers, top performers, and the cumulative dominance of large institutions. This structured approach ensures data quality and clarity in insights.